

Program : Diploma in Tool & Die Engineering	
Course Code : 3113	Course Title: Tool Engineering Drawing
Semester : 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0 T:0, P:3)	Periods per semester: 45

Course Objectives:

- To provide a set of rules regarding the dimensions and tolerances of mating machined parts
- To give the standard procedure of indicating surface texture and its application on drawing
- To explain the forms and proportion of screw threads & fasteners
- To explain the importance of geometrical tolerance
- To prepare assembly views and to create the detail drawing of its elements

Course Prerequisites:

Topic	Course code	Course name	Semester
Basics of Engineering graphics		Engineering Graphics	1

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate different type of Fasteners.	9	Applying
CO2	Prepare dimensional drawings with tolerances and standard procedure of surface texture.	9	Applying
CO3	Apply the concept of sectional views to depict assembly views and detailed drawings of Machine elements.	12	Applying
CO4	Sketch assembly views and detailed drawings of Jig, Fixture & Dies.	12	Applying
	Series Test	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO
CO1	3						
CO2	3						
CO3	3						3
CO4	3						3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate different type of Fasteners		
M1.01	Illustrate various types of screw thread profiles.	3	Applying
M1.02	Sketch different types of bolts and nuts.	3	Applying
M1.03	Illustrate various types of rivet heads.	3	Applying
Contents: Screw thread nomenclature. Major dia, minor dia, pitch, lead, flank, crest, root, thread angle etc. Forms of thread - metric thread, V thread, BSW thread, American national thread, Butress thread, Square thread, ACME thread, and worm thread. Single and multi-start thread. Thread specification & Conventional representation of threads. Bolt, nut, set screws, locking screws, lock washers, nuts. Socket head cap screw, Allen screw, grub screw and CSK screw and assembly, foundation bolts etc. Draw riveted joints Calculate the pitch and arrangements of rivet in rows (Lap and Butt joints).			
CO2	Prepare dimensional drawings with tolerances and standard procedure of surface texture		
M2.01	Compute the tolerances for various types of fits using standard tables	3	Applying
M2.02	Interpret various Surface texture symbols from drawings	3	Applying
M2.03	Apply Geometrical Tolerances in drawings	3	Applying
	Series Test – I	1.5	

Contents:

Classifications of dimensions- functional, nonfunctional, auxiliary and features.

IS specified tolerance. – Indian standard/ISO system of limits and fits (IS919, ISO 286) – Allowances

Basic size deviations and tolerance.

Method of placing limit dimensions. – Bilateral, unilateral.

Fits. – Clearance fit, transition and interference fit.

General tolerance. – IS 2102 {part1 and part2}.

Compute fit and tolerance from tables.

Introduction, - Types of surface texture, waviness nomenclature of surface texture.

Indication of surface roughness.

Symbols of specifying the direction of lay.

Machining symbols.

Introduction about geometrical tolerance.

Tolerance frame and features

Datum. – Primary, secondary and tertiary datum

CO3	Apply the concept of sectional views to depict assembly views and detailed drawings of Machine elements.		
M3.01	Prepare Sectional elevations, side views and plan of various machine parts.	3	Applying
M3.02	Introduce assembly and part drawing	9	Applying

Contents:

Review of sectioning – Conventions showing the section – symbolic representation of cutting plane- types of section – full section, half section, offset section, revolved section, broken section, removed section – section lining

General rules of sectioning. - hatching of single object, adjacent object, thin material, large area, rib & web section.

Detailed drawings of following machine parts are given to students to assemble and draw the sectional or plain elevations / plans / and side views with dimensioning and bill of materials.

1. Screw Jack
2. C Clamp
3. Tool Makers clamp
4. Lathe Tailstock
5. Lathe spindle
6. Bench Vice etc.

CO4	Sketch assembly views and detailed drawings of Jig, Fixture & Dies.		
M4.01	Prepare assembly drawings of production devices like Jigs, Fixtures, Press Tools and Moulds with GD & T, surface roughness values and bill of materials.	12	Applying
	Series Test – II	1.5	
Contents: Following detailed drawings are given to students to assemble and draw the sectional or plain elevations / plans / and side views with dimensioning and bill of materials. Drill Jig – post & latch type Welding fixture, milling fixture, indexing fixture Press tool assembly – Blanking tool Plastic Moulding tool assembly – Injection moulding.			

Text/ Reference:

T/R	Book Title/Author
T1	Machine drawing - P I Varghese and K.C John
R1	Engineering Drawing - N.D. Bhatt
R2	Jigs and Fixtures – Hiram.E.Grant
R3	Jigs and Fixtures Design Manual –P H Joshi
R4	Jigs & Fixtures - P H Joshi

Online Resources:

Sl.No	Website Link
1	http://www.scce.ac.in/noticeboard/10367_23062016pdp-complete-manual.pdf
2	http://www.ignou.ac.in/upload/jig.pdf